

Ganesh M

AWS DevOps Engineer

AWS Certified Solutions Architect

✉ ganeshmkumar.smmkg@gmail.com

☎ +91 8015 147 595

🌐 www.linkedin.com/in/ganesh-m-devops

📍 Chennai, India

PROFESSIONAL SUMMARY

AWS Certified Solutions Architect with 10+ years of IT industry experience, including 6.5+ years of dedicated AWS DevOps engineering expertise in cloud infrastructure automation, CI/CD pipelines, container orchestration, and Site Reliability Engineering (SRE) practices. Proven expertise delivering enterprise-grade CI/CD pipelines, Infrastructure as Code (IaC), and container orchestration solutions for clients in financial services (MUFG Bank), insurance (Direct Line Group, UK), and e-commerce (Groupon, Home Depot). Skilled in Terraform, Kubernetes (EKS), Docker, Jenkins, Ansible, and the full AWS DevOps toolchain. Experienced in observability stack design (Prometheus, Grafana, CloudWatch), and microservices architecture migration. Passionate about driving cloud-native transformation and operational excellence.

KEY ACHIEVEMENTS

- ▶ Reduced deployment cycle time by 30% by designing and optimising multi-stage AWS CodePipeline CI/CD workflows for Direct Line Group (UK), eliminating manual release bottlenecks.
- ▶ Achieved 99.9%+ infrastructure uptime by implementing comprehensive CloudWatch observability with custom alarms, dashboards, and automated incident runbooks.
- ▶ Automated 100% of AWS infrastructure provisioning using Terraform IaC, eliminating configuration drift and enabling repeatable, version-controlled deployments across all environments.
- ▶ Deployed production Kubernetes (EKS) clusters with Grafana-based real-time monitoring dashboards, reducing Mean Time to Resolution (MTTR) for Translink Northern Ireland.
- ▶ Strengthened application security posture by deploying AWS WAF (WebACL) on Application Load Balancers, protecting against OWASP Top 10 threats and DDoS attacks.
- ▶ Led on-premise to AWS cloud migration using Docker containerisation and Kubernetes orchestration, improving scalability and reducing infrastructure overhead.

TECHNICAL SKILLS

Cloud Platforms: Amazon Web Services (AWS) — EC2, VPC, IAM, S3, Lambda (Serverless), ECS, EKS, ECR, RDS (PostgreSQL, MySQL), Route 53, CloudFront, SNS, SQS, CloudFormation, SSM, MWAA (Apache Airflow), EventBridge, WebACL, Glue, ElastiCache

Infrastructure as Code (IaC): Terraform (modules, workspaces, remote state, Terragrunt), AWS CloudFormation (nested stacks, StackSets), Ansible (playbooks, roles, ad-hoc commands, dynamic inventory)

CI/CD & GitOps: Jenkins (pipelines-as-code, shared libraries), AWS CodePipeline, CodeBuild, CodeDeploy, CodeCommit, GitHub Actions, GitLab CI, GitOps workflows, ArgoCD (familiar), Maven, SonarQube (SAST), Nexus

Containers & Orchestration: Docker (multi-stage builds, private registry, Docker Swarm), Kubernetes (EKS, Pods, Services, Deployments, ConfigMaps, Secrets, RBAC, Helm charts, Horizontal Pod Autoscaler), Amazon ECS (Fargate & EC2 launch types)

Observability & Monitoring: Prometheus, Grafana (dashboards, alerting), AWS CloudWatch (custom metrics, log insights, alarms), AWS CloudTrail, AppDynamics APM, ELK Stack (Elasticsearch, Logstash, Kibana — familiar).

Security & Compliance: AWS IAM (roles, policies, least-privilege), AWS WAF / WebACL, SSL/TLS certificate management, AWS Security Hub (familiar), OWASP Top 10 remediation.

Networking & Architecture: VPC design (subnets, routing tables, NACLs, security groups, VPC peering, Transit Gateway), Application Load Balancer (ALB), Network Load Balancer (NLB), DNS (Route 53, health checks, failover routing), API Gateway, CDN (CloudFront)

Scripting & Programming: Python (automation scripts, Lambda functions, Boto3 SDK), Bash/Shell scripting, YAML, JSON, HCL (Terraform)

Databases & Integration: Amazon RDS (MySQL, PostgreSQL), DynamoDB (NoSQL), ElastiCache (Redis), MuleSoft Anypoint Platform (enterprise API integration)

Web Servers & OS: Apache HTTP Server, Apache Tomcat, Linux (Ubuntu, Amazon Linux 2, RHEL — administration, hardening), Windows Server

Agile & Collaboration: Agile / Scrum methodology, ITIL incident & change management, Jira, Confluence, GitHub, Cross-functional team collaboration

CERTIFICATIONS

AWS Certified Solutions Architect – Associate · Amazon Web Services June 2024 – June 2027

In progress / Planned: AWS Certified DevOps Engineer – Professional, Certified Kubernetes Administrator (CKA)

PROFESSIONAL EXPERIENCE

AWS DevOps Engineer | *Fractal Analytics*

June 2023 – Present · Chennai, India

► **Client: Direct Line Group (UK) — Financial Services & Insurance**

- Managed and monitored end-to-end AWS infrastructure (EC2, S3, RDS, VPC) for a UK-based Tier 1 insurance provider, maintaining 99.9%+ uptime SLAs.
- Designed multi-stage CI/CD pipelines using AWS CodePipeline, CodeBuild, and CodeDeploy, reducing deployment cycle time by 30% and eliminating manual release processes.
- Authored Terraform modules for all AWS infrastructure provisioning — VPC, EC2, RDS, IAM roles — enabling version-controlled, zero-drift, repeatable deployments across dev, staging, and production environments.
- Automated cloud infrastructure tasks using AWS Systems Manager (SSM) Run Command and State Manager, reducing manual operational toil by 60%.
- Built AWS Lambda functions integrated with CodePipeline to send real-time deployment status and failure alerts to Microsoft Teams, improving engineering team response time.
- Configured CloudWatch custom metrics, log-based alarms, and operational dashboards to achieve proactive incident detection and sub-5-minute alerting response.
- Installed and configured MuleSoft Anypoint Platform for enterprise API integration, enabling seamless data exchange between internal microservices and third-party insurance systems.
- Optimised application performance and availability using AppDynamics APM; identified and resolved three critical Java heap memory bottlenecks reducing P99 latency by 25%.

► **Client: Translink (Northern Ireland) — Public Transport Operator**

- Deployed and managed production Grafana on Amazon EKS, integrating with CloudWatch and custom Lambda-published metrics to provide real-time visibility into 50+ KPIs across infrastructure and application layers.
- Built 12 Grafana dashboards from scratch covering infrastructure health, deployment frequency, error rates, and SLO tracking for development and operations teams.
- Developed Docker-based microservices and deployed containerised workloads to Amazon ECS (Fargate), enabling auto-scaling and cost-efficient compute utilisation.
- Implemented Kubernetes resources on AWS EKS — Deployments, Services, ConfigMaps, Secrets, RBAC policies, and Horizontal Pod Autoscalers — managing 20+ containerised application workloads.
- Enforced application security by deploying AWS WAF with custom WebACL rule groups on Application Load Balancers, mitigating SQL injection, XSS, and rate-limit attack vectors.
- Automated full AWS infrastructure stack provisioning with Terraform workspaces (dev/staging/prod), cutting environment spin-up time from days to under 30 minutes.

DevOps Engineer | *Tata Consultancy Services (TCS)*

Apr 2022 – June 2023 · India

► **Client: MUFG Bank — Global Financial Institution**

- Owned and operated end-to-end CI/CD pipeline lifecycle using Jenkins declarative pipelines for a critical global banking application serving millions of users across APAC and EMEA.
- Integrated SonarQube static code analysis (SAST) into Jenkins pipelines, enforcing quality gates and reducing code vulnerability findings by 40%.
- Managed Git branching strategy (GitFlow) across 15+ repositories, including feature, release, and hotfix branch governance and merge request review workflows.
- Authored Ansible playbooks and roles for automated configuration management, provisioning, and application deployment across 30+ development and staging servers.
- Led AWS cloud infrastructure build-out using Terraform, including VPC, subnets, security groups, EC2 Auto Scaling groups, and RDS instances for development and UAT environments.
- Executed on-premise to AWS cloud migration for legacy Java application workloads, using Docker containerisation and Amazon ECS to achieve cloud-native architecture.
- Resolved 95%+ of CI/CD pipeline failures within SLA by diagnosing build script inconsistencies, dependency version conflicts, and environment configuration issues.

Cloud & DevOps Engineer | *Groupon Shared Services Pvt Ltd*

July 2020 – Mar 2022 · India

► **Client: Home Depot (USA) — Retail & E-Commerce**

- Architected three-tier AWS infrastructure (web, application, data layers) using EC2 Auto Scaling, ELB, VPC, IAM, S3, and RDS to support a high-traffic food ordering and delivery platform.
- Implemented multi-region DNS routing and traffic management with Route 53 (weighted routing, health checks, failover policies) ensuring 99.95% availability during peak traffic events.
- Developed event-driven monitoring and alerting system using CloudWatch, SNS, and SQS, enabling real-time notification of system anomalies to on-call engineering teams.
- Deployed and managed Apache HTTP Server and Apache Tomcat application servers, configuring virtual hosts, SSL termination, load balancing, and JVM performance tuning. Maintained Git version control workflows and deployed microservices (Spring Boot) to AWS EC2 with zero-downtime rolling deployment strategies using Elastic Load Balancing.

► **Client: Magento (UK) — E-Commerce Platform**

- Provided day-to-day network and application support for a UK e-commerce client in a 24×7 production environment.
- Handled L1/L2 incident management, change requests, and service requests via ticketing system following ITIL processes.
- Managed user access controls, repository permissions, and enforced branch protection policies in GitHub.
- Supported SSL/TLS certificate renewals and basic server configuration tasks.
- Maintained communication with cross-functional teams for incident escalation and resolution.

IT Support Engineer | *Solartis Technology Pvt Ltd*

May 2015 – Apr 2017 · India

- Administered Linux (Ubuntu) and Windows Server environments — user account management, disk/storage management, system performance tuning, and OS-level security hardening.

EDUCATION

Bachelor of Computer Application (BCA)

2012 – 2015

A.M. Jain College, Chennai, Tamil Nadu